



مبانی برنامه نویسی

Introduction to computers and C

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https://github.com/safayani/Programming_Basics_course



Introduction

- Software (the C instructions you write, which are also called **code**)
- Hardware (computers and related devices)

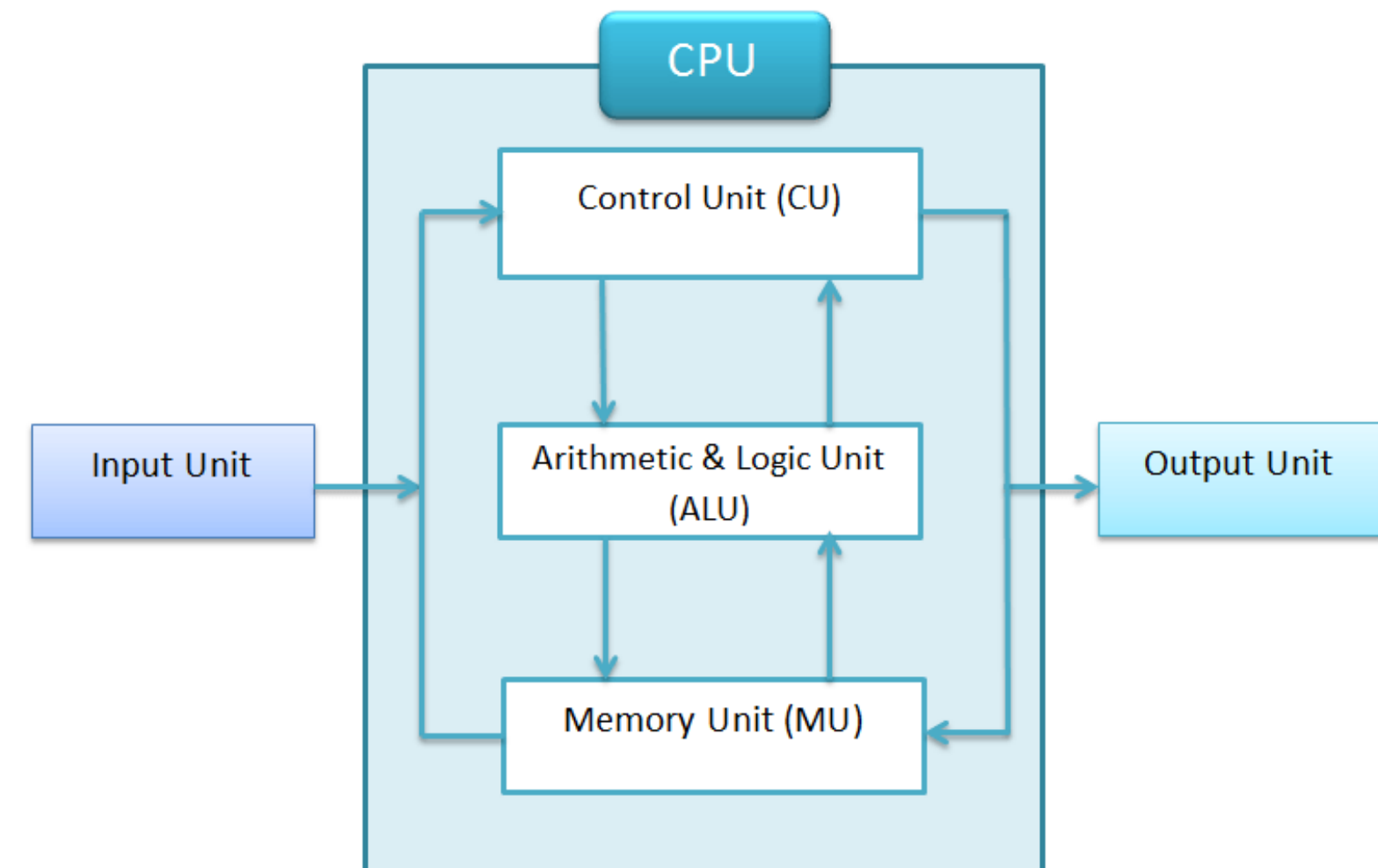
Computer Organization

- Input units (keyboards, touch screens, mice and touchpads, microphone, webcam, ...)
- Output units (Monitor, printer, internet, speaker, ...)
- Memory unit
 - RAM (Random Access Memory)
 - Bit: smallest store space, either 0 or 1
 - Byte: a byte is 8 bit
 - 128 GB of RAM

Unit	Bytes	Which is approximately
1 kilobyte (KB)	1024 bytes	10^3 bytes (1024 bytes exactly)
1 megabyte (MB)	1024 kilobytes	10^6 (1,000,000) bytes
1 gigabyte (GB)	1024 megabytes	10^9 (1,000,000,000) bytes
1 terabyte (TB)	1024 gigabytes	10^{12} (1,000,000,000,000) bytes
1 petabyte (PB)	1024 terabytes	10^{15} (1,000,000,000,000,000) bytes
1 exabyte (EB)	1024 petabytes	10^{18} (1,000,000,000,000,000,000) bytes
1 zettabyte (ZB)	1024 exabytes	10^{21} (1,000,000,000,000,000,000,000) bytes

Computer Organization

- ALU (Arithmetic and logic unit)
 - Add, subtract, multiply, division
 - Make decisions: comparing two items from the memory unit to determine whether they're equal
 - In today's systems, the ALU is part of the CPU



Computer Organization

- CPU (Central Processing Unit)
 - This “administrative” section coordinates and supervises the operation of the other; The CPU tells:
 - the input unit when to read information into the memory unit,
 - the ALU when to use information from the memory unit in calculations
 - the output unit when to send information from the memory unit to specific output devices.
 - A **dual-core processor** has two CPUs, a **quad-core processor** has four and an **octa-core processor** has eight

Computer Organization

- **Secondary Storage Unit**

- Information on secondary storage devices is **persistent**—it's preserved even when the computer's power is turned off.
- SSD(solid-state drives)
- USB flash drives
- HDD (hard disk drive)

Data hierarchy

- **Bits**

- a digit that can assume one of *two* values—and is a computer's smallest data item. It can have the value 0 or 1.

- **Characters**

- Digits, letters and special symbols are known as **characters**.
- **decimal digits** (_{0–9}), **letters** (_{A–Z} and _{a–z}) and **special symbols** such as \$ @ % & * () – + " : ; , ? /
- C uses the **ASCII (American Standard Code for Information Interchange)** character set by default.
- C also supports **Unicode[®]** characters

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

The ASCII Character Set

Machine Languages, Assembly Languages and High-Level Languages

- Machine Languages:
- Assembly Languages and Assemblers
- High-Level Languages and Compilers

Machine Languages, Assembly Languages and High-Level Languages

Machine Code	Assembly Code	C code
B8 05 00 00 00 ; mov eax, 5 BB 03 00 00 00 ; mov ebx, 3 01 D8 ; add eax, ebx B8 01 00 00 00 ; mov eax, 1 CD 80 ; int 0x80	section .text global _start _start: mov eax, 5 ; Store 5 in register eax mov ebx, 3 ; Store 3 in register ebx add eax, ebx ; Add ebx to eax (eax = eax + ebx) mov eax, 1 ; System call for exit int 0x80 ; Call kernel	int main() { int a = 5; int b = 3; int result = a + b; return result; }

Operating Systems

- **Windows**

- In the mid-1980s, Microsoft developed the **Windows operating system**, consisting of a graphical user interface built on top of DOS (Disk Operating System)

- **Linux—An Open-Source Operating System**

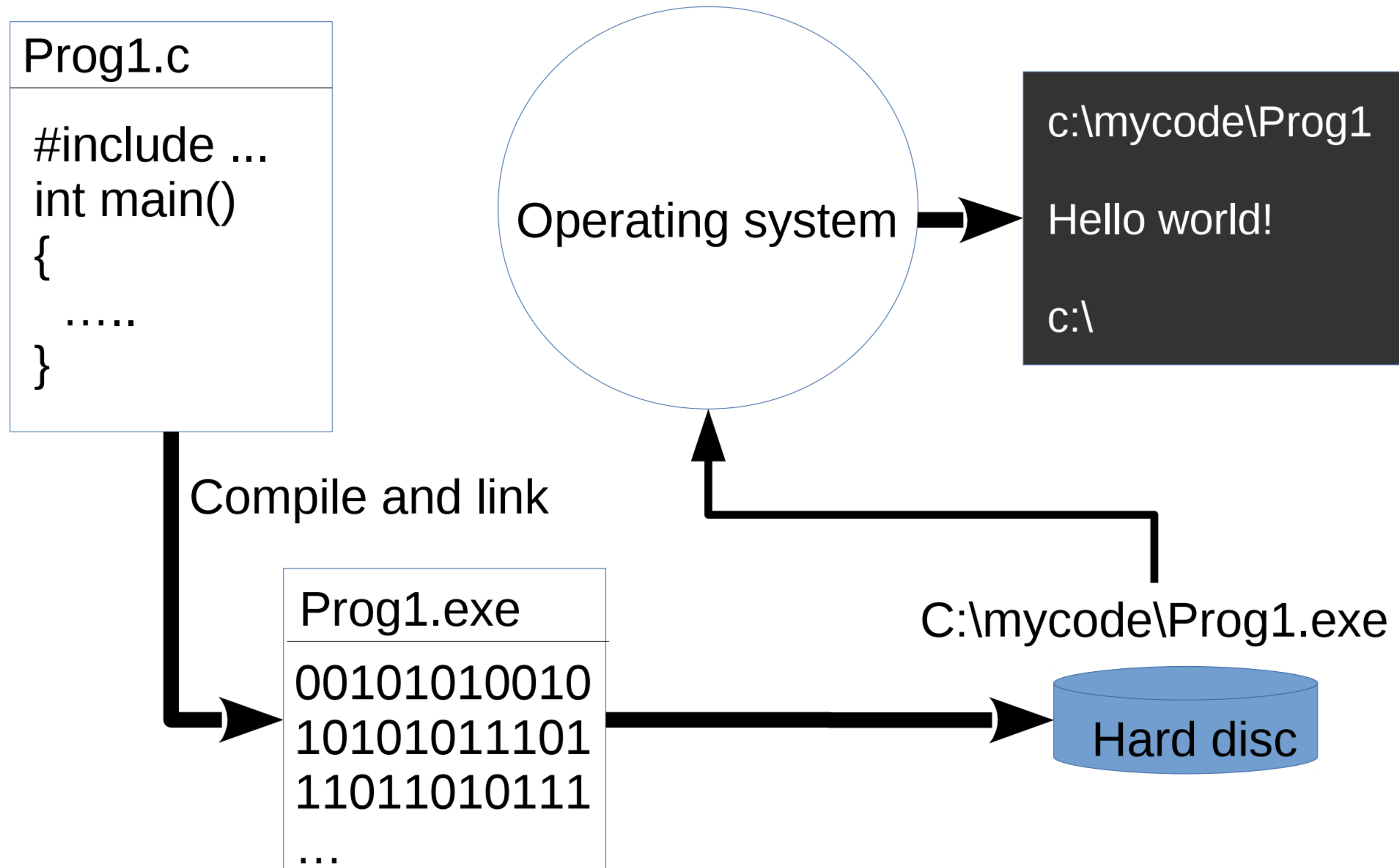
- **Apple's macOS and Apple's iOS for iPhone® and iPad® Devices**

- **Google's Android**

The C Programming Language

- The C language was evolved from B by Dennis Ritchie at Bell Laboratories and was originally implemented in 1972
- **Built for Performance**
 - C is widely used to develop systems that demand performance, such as operating systems, embedded systems, real-time systems and communications systems
 - Today's popular desktop operating systems—**Windows, macOS and Linux**—are partially written in C.
 - Many popular applications are partially written in C, including popular web browsers (**e.g., Google Chrome and Mozilla Firefox**), database management systems (**e.g., Microsoft SQL Server, Oracle and MySQL**) and more.

The life-time of a computer program



Basics of a Typical C Program Development Process

- **Phases of C Programs:**

1. Edit

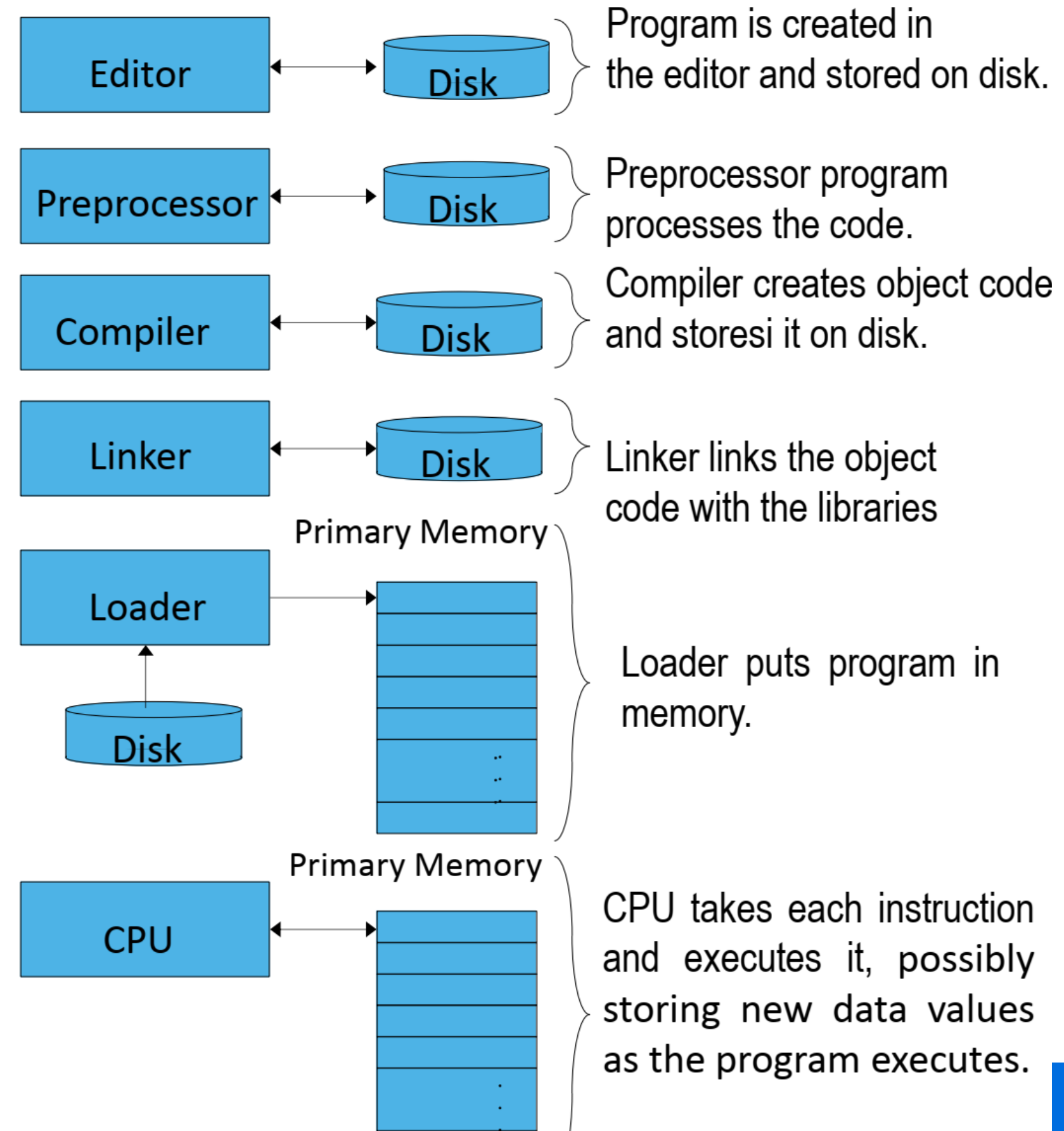
2. Preprocess

3. Compile

4. Link

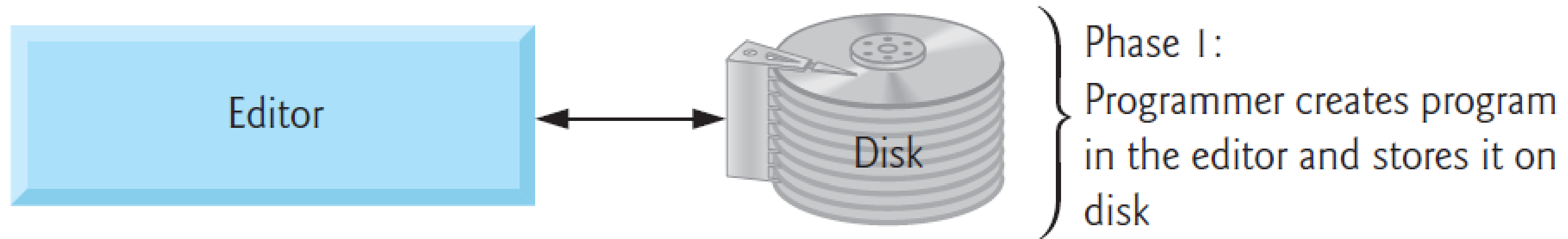
5. Load

6. Execute

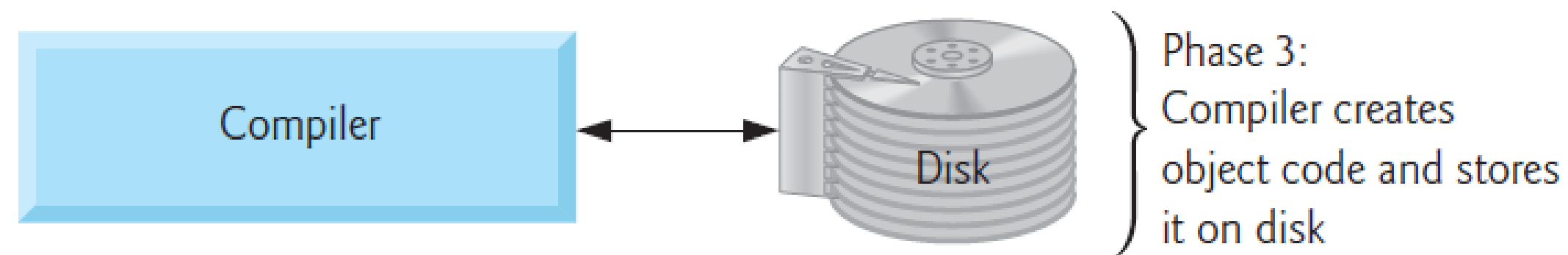


C Program development

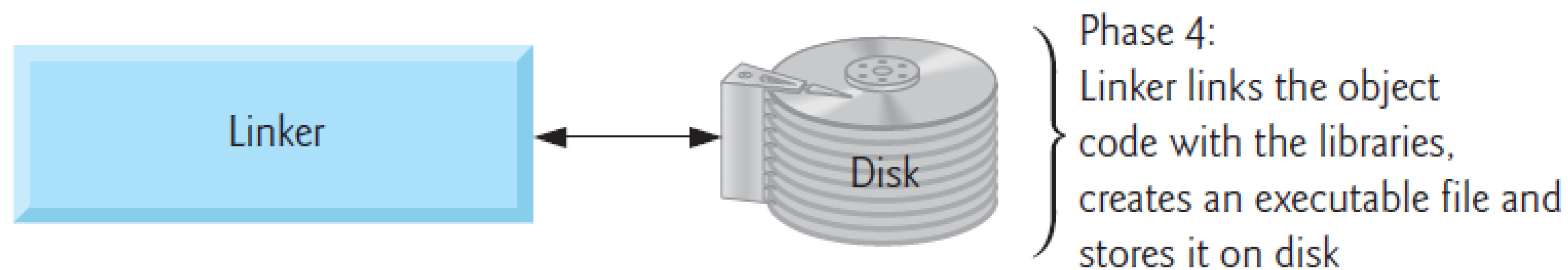
- **Phase 1: Creating a Program**



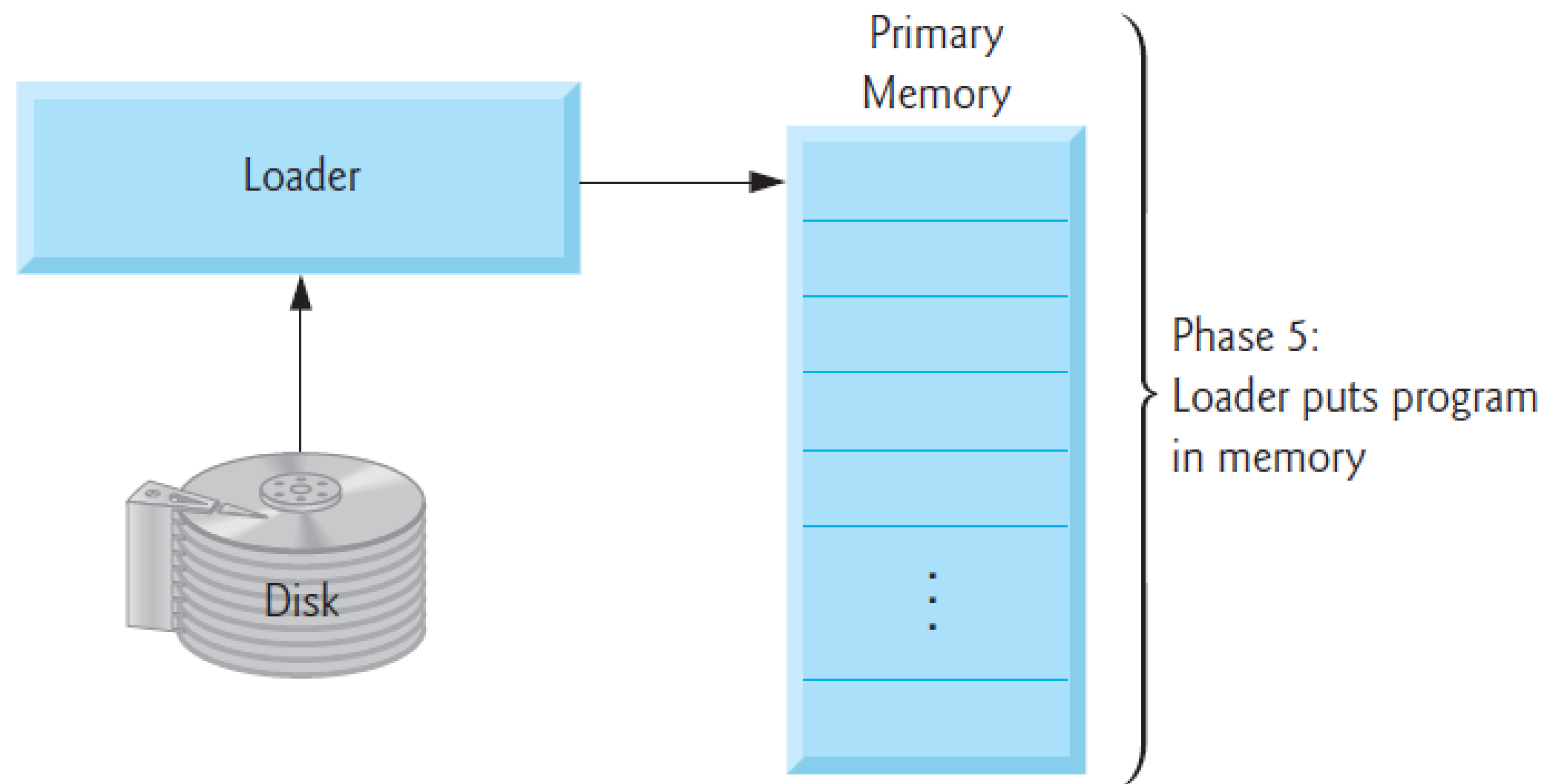
- Phase 2 and 3: Preprocessing and Compiling a C Program:
The compiler translates the C program into machine-language code (also referred to as **object code**).



- Linking: C programs typically use functions defined elsewhere, such as in the standard libraries, open-source libraries or private libraries of a particular project. The object code produced by the C compiler typically contains “holes” due to these missing parts. A **linker** links a program’s object code with the code for the missing functions to produce an **executable image** (with no missing pieces).



- Phase 5: Loading



- Phase 6: Execution

